Question			Answer	Marks	AO	Guidance
15	(a)		$\frac{dy}{dx} = 2ax + b$	M1	1.1	Allow $2at + b$
			$y - (at^2 + bt + c) = (2at + b)(x - t)$ **or** $at^2 + bt + c = (2at + b)t + k$	M1	1.1	Use of a form of equation of straight line with *their* gradient and $x = t$
			$y = (2at + b)x - 2at^2 - bt + at^2 + bt + c$ So $\boldsymbol{y = (2at + b)x - at^2 + c}$	A1	2.1	Convincing completion to correct result
			Alternative method			
			The line and curve cross where $\boldsymbol{ax^2 + bx + c = (2at + b)x - at^2 + c}$	M1		
			$\boldsymbol{ax^2 - 2atx + at^2 = 0}$	M1		Getting everything to one side
			$a(x - t)^2 = 0$ Line touches curve when $x = t$ so it is the required tangent	A1		Convincing completion to correct result
				[3]		
15	(b)		Tangents cross where $\boldsymbol{(2ax_P + b)x - ax_P^2 + c = (2ax_Q + b)x - ax_Q^2 + c}$	M1	2.1	Use of tangent formula with distinct values of t Condone alternative notation for co-ordinates. e.g. P and Q, m and n, and other subscript versions
			$\boldsymbol{(2ax_P - 2ax_Q)x = ax_P^2 - ax_Q^2}$	M1	1.1	Getting terms in x on one side
			$x = \frac{x_P^2 - x_Q^2}{2(x_P - x_Q)}$ $= \frac{(x_P + x_Q)(x_P - x_Q)}{2(x_P - x_Q)} = \frac{x_P + x_Q}{2}$	A1	2.1	Convincing completion to given result. Accept the final result in terms of *their* variables.
				[3]		

Q15a. The first M1 is for differentiating. We are allowing the gradient as $2at + b$ as well as $2ax + b$. Some are just writing $2ax+b/2at+b$ and that isn't convincing that they have differentiated – rather just copying from the given answer so give M0.
The second M1 is for substituting into the equation of a straight line using their gradient and $x = t$. They could use $(y - y_1) = m(x - x_1)$ or $y= mx + c$. Condone a lack of brackets around their m if recovered.
The final A1 is for getting to the correct answer. As the answer is given, we need to see some working here.

Q15b. The first M1 is for equating their two tangents with distinct values of t. These could be x_p and x_q as per the scheme, but they might choose other pairs of symbols – e.g. P and Q, m and n, t_1 and t_2.
The second M1 is for getting the x terms on one side and the non-x terms on the other.
To get the A1 they must show the working out including the difference of two squares and cancelling to get the required result. Allow the final result in terms of their variable.